



COMPETENCY STANDARD

FOR

**APPLIED SEMICONDUCTOR SYSTEMS: POWER
DEVICES AND INDUSTRIAL INTEGRATION**

(SEMICONDUCTOR / VLSI SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Applied Semiconductor Systems: Power Devices and Industrial Integration serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the Semiconductor/ VLSI sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for Applied Semiconductor Systems: Power Devices and Industrial Integration. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.

- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (08 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-ASS-01-G	Apply OHS, ESD and Safe Workplace Practices	1. Identify OHS and ESD requirements 2. Practice safe workstation and EDA lab procedures 3. Follow emergency response procedures	04
SICIP-SCD-ASS-02-G	Apply Workplace Communication and Professional Ethics	1. Interpret workplace communication and etiquette 2. Prepare technical documents 3. Participate in design review and presentation	04
Total Hour			08 Hrs.

Sector Specific Competencies (32 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-ASS-01-S	Interpret Power Semiconductor Systems and Industrial Applications	1. Explain power semiconductor devices. 2. Identify industrial applications. 3. Interpret device ratings and datasheets. 4. Recognize power electronics systems.	08
SICIP-SCD-ASS-02-S	Apply Power Electronics and Simulation Fundamentals	1. Explain power electronics fundamentals 2. Interpret PWM control concepts. 3. Perform circuit simulation. 4. Operate measurement instruments safely.	08
SICIP-SCD-ASS-03-S	Apply Embedded Control and Sensor Integration	1. Interface power stages with MCUs. 2. Configure control signals. 3. Implement embedded control functions. 4. Apply protection techniques.	08
SICIP-SCD-ASS-04-S	Prepare Power System Design Documentation	1. Prepare power electronic schematics. 2. Interpret PCB layout requirements. 3. Prepare bills of materials (BOM). 4. Prepare test documentation and reports.	08
Total Hour			32 Hrs.

Occupation Specific Competencies (200 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-ASS-01-O	Apply Power Semiconductor Devices and Switching Techniques	1. Identify power semiconductor devices. 2. Analyze switching characteristics and losses. 3. Select devices for safe and efficient operation. 4. Simulate MOSFET and IGBT characteristics.	32
SICIP-SCD-ASS-02-O	Implement Gate Drive Circuits and Thermal Management	1. Design gate driver circuits. 2. Apply gate drive techniques. 3. Configure protection and dead-time. 4. Select thermal management solutions.	32
SICIP-SCD-ASS-03-O	Design Power Conversion Systems	1. Develop DC-DC converter topologies. 2. Explain inverter and UPS systems. 3. Select magnetic and filter components. 4. Verify converter performance.	40
SICIP-SCD-ASS-04-O	Integrate Control, Motor Drive and Renewable Energy Systems	1. Interface power stages with MCUs. 2. Implement motor control functions. 3. Integrate BMS, solar inverter and MPPT systems. 4. Validate industrial control applications.	36
SICIP-SCD-ASS-05-O	Perform Circuit Design, PCB Development and Troubleshooting	1. Prepare power electronic schematics. 2. Develop PCB layouts. 3. Assemble and prototype circuits. 4. Diagnose circuit faults. 5. Mitigate EMI/EMC issues.	32
SICIP-SCD-ASS-06-O	Develop Semiconductor Systems Integration Project	1. Explain semiconductor packaging and reliability. 2. Plan the system integration project. 3. Develop and optimize the prototype. 4. Test system performance. 5. Prepare technical documentation.	28
Total Hour			000 Hrs.

COMPETENCY STANDARD: APPLIED SEMICONDUCTOR SYSTEMS: POWER DEVICES AND INDUSTRIAL INTEGRATION

The Generic Competencies

Unit of Competency: APPLY OHS, ESD AND SAFE WORKPLACE PRACTICES	Nominal Duration: 04 hrs.	Unit Code: SICIP-SCD-ASS-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply OHS, ESD and Safe Workplace Practices. It specifically includes the tasks of identifying OHS and ESD requirements, practicing safe workstation and EDA lab procedures and following emergency response procedures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS and ESD requirements	1.1 <u>Workplace OHS and ESD requirements</u> are identified according to organizational procedures. 1.2 <u>Hazards</u> associated with semiconductor and EDA laboratory environments are recognized. 1.3 <u>Personal protective equipment (PPE) and ESD protection devices</u> are identified and used appropriately. 1.4 ESD control measures are applied to protect electronic components and equipment. 1.5 <u>Workplace safety signs, symbols, and instructions</u> are interpreted correctly.
2. Practice safe workstation and EDA lab procedures	2.1 <u>Workstations</u> are prepared according to OHS and ESD requirements. 2.2 <u>EDA tools, computers, and laboratory equipment</u> are operated following safe work procedures. 2.3 <u>Cables, electrical equipment, and work areas</u> are maintained to prevent workplace hazards. 2.4 Good housekeeping practices are maintained to ensure a safe working environment. 2.5 Workplace incidents, hazards, and unsafe conditions are identified and reported promptly.
3. Follow emergency response procedures	3.1 <u>Emergency procedures</u> are identified according to workplace requirements. 3.2 Emergency exits, assembly points, and safety equipment are located and accessed correctly. 3.3 Appropriate actions are taken during electrical, fire, or other workplace emergencies. 3.4 Incidents and accidents are reported according to

	organizational procedures.
	3.5 <u>Emergency response activities</u> are carried out in accordance with workplace safety requirements.

Range of Variables

Variable	Range May include but not limited to:
1. Workplace OHS and ESD requirements	1.1 OHS Policies 1.2 ESD Procedures 1.3 Workplace Rules 1.4 Safe Work Practices 1.5 Emergency Procedures 1.6 Risk Assessment
2. Hazards	2.1 Electrostatic Discharge (ESD) 2.2 Electrical Hazards 2.3 Fire Hazards 2.4 Chemical Hazards 2.5 Trip and Slip Hazards 2.6 Equipment Hazards
3. Personal protective equipment (PPE) and ESD protection devices	3.1 ESD Wrist Strap 3.2 ESD Shoes 3.3 ESD Mat 3.4 Safety Glasses 3.5 Protective Gloves 3.6 Lab Coat
4. Workplace safety signs, symbols, and instructions	4.1 Warning Signs 4.2 Mandatory Signs 4.3 Prohibition Signs 4.4 Emergency Exit Signs 4.5 Fire Safety Signs 4.6 Hazard Symbols
5. Workstations	5.1 ESD Workstation 5.2 Computer Workstation 5.3 Design Workbench 5.4 Laboratory Bench 5.5 Test Workstation 5.6 Simulation Workstation
6. EDA tools, computers, and laboratory equipment	6.1 EDA Software 6.2 Workstation Computer 6.3 Server 6.4 Oscilloscope 6.5 Digital Multimeter 6.6 Power Supply
7. Cables, electrical equipment, and work areas	7.1 Power Cables 7.2 Network Cables 7.3 Extension Cords

	7.4 Electrical Outlets 7.5 Work Benches 7.6 Cable Management
8. Emergency procedures	8.1 Fire Emergency 8.2 Electrical Emergency 8.3 Medical Emergency 8.4 Evacuation Procedure 8.5 First Aid Procedure 8.6 Incident Reporting Procedure
9. Emergency response activities	9.1 Evacuation 9.2 First Aid 9.3 Firefighting 9.4 Hazard Isolation 9.5 Emergency Notification 9.6 Incident Documentation

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Principles of Occupational Health and Safety (OHS) and Electrostatic Discharge (ESD) protection. 1.2 Common hazards in semiconductor fabrication, EDA laboratories, and electronic work environments. 1.3 Proper use of Personal Protective Equipment (PPE) and ESD protection devices. 1.4 ESD control methods and safe handling of electronic components. 1.5 Safe operation of computers, EDA tools, electrical equipment, and laboratory workstations. 1.6 Workplace housekeeping practices and electrical safety requirements. 1.7 Emergency procedures, including fire safety, electrical hazards, evacuation routes. 1.8 Interpretation of workplace safety signs, symbols, and organizational safety policies and procedures.
2. Underpinning Skills	2.1 Applying workplace OHS and ESD requirements in semiconductor and EDA laboratory environments 2.2 Interpreting workplace safety signs, symbols, and safety instructions 2.3 Preparing and maintaining safe workstations and EDA laboratory environments 2.4 Operating EDA tools, computers, and laboratory equipment safely 2.5 Applying good housekeeping practices to maintain a hazard-free workplace 2.6 Following emergency response procedures for electrical, fire, and workplace emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety

	3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning/operational manual 4.3 PPEs 4.4 Firefighting equipment 4.5 Emergency response manual 4.6 Emergency Systems (Fire alarms, emergency exits, etc.) 4.7 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified OHS and ESD requirements 1.2 practiced safe workstation and EDA lab procedures 1.3 followed emergency response procedures
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY WORKPLACE COMMUNICATION AND PROFESSIONAL ETHICS	Nominal Duration: 04 hrs.	Unit Code: SICIP-SCD-ASS-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply workplace communication and professional ethics. It specifically includes the tasks of interpreting workplace communication and etiquette, preparing technical documents and participating in design review and presentation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 <u>Workplace communication methods</u> and organizational protocols are identified. 1.2 <u>Professional etiquette</u> is demonstrated during workplace interactions. 1.3 Technical information and work instructions are interpreted accurately. 1.4 <u>Verbal and written communication</u> is conducted clearly and professionally. 1.5 Team communication and collaboration are maintained according to workplace requirements.
2. Prepare technical documents	2.1 <u>Technical information</u> is collected and organized according to project requirements. 2.2 Design notes, reports, and technical documents are prepared using standard formats. 2.3 <u>Technical terminology and documentation standards</u> are applied correctly. 2.4 Documents are reviewed for accuracy, completeness, and consistency. 2.5 Final documents are submitted according to organizational procedures.
3. Participate in design review and presentation	3.1 <u>Design data and supporting materials</u> are prepared for review meetings. 3.2 Design concepts and technical information are presented clearly to stakeholders. 3.3 Feedback and review comments are recorded and analyzed. 3.4 <u>Design modifications</u> are incorporated based on review outcomes. 3.5 Professional communication and teamwork are demonstrated throughout the review and presentation process.

Range of Variables

Variable	Range May Include but not limited to:
1. Workplace communication methods	1.1 Face-to-Face Communication 1.2 Email 1.3 Phone Calls 1.4 Video Conferencing 1.5 Instant Messaging
2. Professional etiquette	2.1 Respectful Behaviour 2.2 Professional Language 2.3 Punctuality 2.4 Active Listening 2.5 Appropriate Dress Code 2.6 Workplace Ethics
3. Verbal and written communication	3.1 Meetings 3.2 Emails 3.3 Reports 3.4 Presentations 3.5 Memos 3.6 Technical Documentation
4. Technical information	4.1 Design Specifications 4.2 Project Requirements 4.3 Technical Drawings 4.4 Test Results 4.5 Design Data 4.6 Reference Documents
5. Technical terminology and documentation standards	5.1 Technical Terms 5.2 Industry Standards 5.3 Naming Conventions 5.4 Document Templates 5.5 File Formats 5.6 Documentation Guidelines
6. Design data and supporting materials	6.1 Design Drawings 6.2 Schematics 6.3 Presentation Slides 6.4 Technical Reports 6.5 Simulation Results 6.6 Design Specifications
7. Design modifications	7.1 Design Updates 7.2 Specification Changes 7.3 Layout Revisions 7.4 Documentation Updates 7.5 Error Corrections 7.6 Improvement Actions

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Principles of workplace communication, professional etiquette, and organizational communication protocols. 1.2 Methods for interpreting technical information, work instructions, and engineering documentation. 1.3 Standards and formats for preparing technical reports, design notes, and project documentation. 1.4 Use of technical terminology, documentation conventions, and version control practices. 1.5 Principles of effective presentation, technical discussion, and design review processes. 1.6 Methods for receiving, incorporating technical feedback into design improvements. 1.7 Teamwork, collaboration, and professional communication practices in engineering.
2. Underpinning Skills	2.1 Applying professional workplace communication and organizational etiquette 2.2 Collaborating with team members to support design and project activities 2.3 Preparing technical documents, reports, and design notes using standard formats 2.4 Applying technical terminology and documentation standards correctly 2.5 Reviewing technical documents for accuracy, completeness, and consistency 2.6 Presenting design concepts and technical information clearly during review meetings 2.7 Incorporating feedback to improve design quality and project outcomes
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Work books 4.3 Learning manuals 4.4 Technical documents

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted workplace communication and etiquette 1.2 prepared technical documents 1.3 participated in design review and presentation
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

Sector Specific Competencies

Unit of Competency: INTERPRET POWER SEMICONDUCTOR SYSTEMS AND INDUSTRIAL APPLICATIONS	Nominal Duration: 08 hrs.	Unit Code: SICIP-SCD-ASS-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret Semiconductor, Systems and Industrial Applications. It specifically includes the tasks of explaining power semiconductor devices, identifying industrial applications, interpreting device ratings and datasheets and recognizing power electronics systems.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain power semiconductor devices.	1.1 <u>Power semiconductor devices and their operating principles</u> are identified. 1.2 Characteristics of diodes, MOSFETs, IGBTs, and thyristors are explained. 1.3 Device operating modes and switching behavior are interpreted. 1.4 Power semiconductor devices are selected according to application requirements. 1.5 Device characteristics are related to industrial power electronic applications.
2. Identify industrial applications.	2.1 <u>Industrial applications of power semiconductor devices</u> are identified. 2.2 Power conversion systems used in industrial equipment are recognized. 2.3 Motor drives, renewable energy systems, electric vehicles, and power supplies are distinguished according to their applications. 2.4 Power semiconductor devices are matched with appropriate industrial applications. 2.5 Industrial application requirements are interpreted according to system specifications.
3. Interpret device ratings and datasheets.	3.1 <u>Power semiconductor datasheets</u> are interpreted according to manufacturer specifications. 3.2 Electrical, thermal, and switching ratings are identified from datasheets. 3.3 Safe operating limits and application conditions are interpreted. 3.4 <u>Device parameters</u> are evaluated for specific

	application requirements. 3.5 Device selection is verified using datasheet information.
4. Recognize power electronics systems.	4.1 <u>Power electronics system components</u> are identified according to system requirements. 4.2 <u>Power conversion systems</u> are recognized. 4.3 Power semiconductor devices are related to power converter operation. 4.4 System block diagrams and power flow are interpreted correctly. 4.5 Power electronics systems are evaluated according to industrial applications.

Range of Variables

Variable	Range May include but not limited to:
1. Power semiconductor devices and their operating principles	1.1 Power Diodes 1.2 Power MOSFETs 1.3 IGBTs 1.4 Thyristors (SCRs) 1.5 TRIACs 1.6 Silicon Carbide (SiC) Devices
2 Industrial applications of power semiconductor devices	2.1 Industrial Motor Drives 2.2 Renewable Energy Systems 2.3 Electric Vehicle Systems 2.4 Industrial Power Supplies 2.5 UPS Systems 2.6 Battery Charging Systems
3 Power semiconductor datasheets	3.1 Manufacturer Datasheets 3.2 Product Specifications 3.3 Application Notes 3.4 Technical Manuals 3.5 Package Information 3.6 Reliability Data
4 Device parameters	4.1 On-State Resistance (RDS (on)) 4.2 Collector-Emitter Saturation Voltage (VCE (sat)) 4.3 Gate Charge 4.4 Threshold Voltage 4.5 Reverse Recovery Time 4.6 Switching Energy
5 Power electronics system components	5.1 Power Semiconductor Devices 5.2 Gate Driver Circuits 5.3 Magnetic Components 5.4 Filters 5.5 Control Circuits 5.6 Power Supply Units

6 Power conversion systems	6.1	AC-DC Rectifiers
	6.2	DC-DC Converters
	6.3	DC-AC Inverters
	6.4	AC-AC Converters
	6.5	Switched-Mode Power Supplies (SMPS)
	6.6	Variable Frequency Drives (VFDs)

Curricular Content Guide

1. Underpinning Knowledge	1.1	Fundamentals of power semiconductor devices and operating principles.
	1.2	Characteristics and switching behavior of diodes, MOSFETs, IGBTs, and thyristors.
	1.3	Industrial applications of power semiconductor devices and power conversion systems.
	1.4	Interpretation of device datasheets, ratings, electrical, thermal, and switching parameters.
	1.5	Power electronics system components, converter operation, and power flow concepts.
	1.6	Device selection criteria based on application requirements and system specifications.
2. Underpinning Skills	2.1	Identifying power semiconductor devices and operating principles
	2.2	Selecting power semiconductor devices for specific applications
	2.3	Interpreting electrical, thermal, and switching ratings from datasheets
	2.4	Evaluating device operating limits and safe operating conditions
	2.5	Recognizing industrial applications of power semiconductor devices
	2.6	Understanding power conversion systems and components
	2.7	Relating semiconductor devices with industrial power electronic applications
3 Underpinning Attitudes	3.1	Commitment to occupational health and safety
	3.2	Promptness in carrying out activities
	3.3	Sincere and honest to duties
	3.4	Environmental concerns
	3.5	Eagerness to learn
	3.6	Tidiness and timeliness
	3.7	Respect for rights of peers and seniors in the workplace
	3.8	Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1	Workplace (simulated or actual)
	4.2	Projector

	4.3 Stationery
	4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 explained power semiconductor devices. 1.2 identified industrial applications. 1.3 interpreted device ratings and datasheets. 1.4 recognized power electronics systems.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY POWER ELECTRONICS AND SIMULATION FUNDAMENTALS	Nominal Duration: 08 hrs.	Unit Code: SICIP-SCD-ASS-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Apply Power Electronics and Simulation Fundamentals. It specifically includes the tasks of explaining power electronics fundamentals, interpreting PWM control concepts, performing circuit simulation and operating measurement instruments safely.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain power electronics fundamentals	1.1 <u>Power electronics principles and circuit fundamentals</u> are interpreted according to design requirements. 1.2 Basic power converter topologies and operating principles are identified. 1.3 Voltage, current, power, and efficiency concepts are explained. 1.4 <u>Power electronic components</u> are selected according to circuit requirements. 1.5 Power electronics fundamentals are related to industrial applications.
2. Interpret PWM control concepts.	2.1 <u>PWM principles and switching techniques</u> are interpreted according to control requirements. 2.2 PWM waveforms and duty cycle concepts are explained. 2.3 <u>PWM control methods</u> are related to power converter operation. 2.4 PWM parameters are interpreted according to application requirements. 2.5 PWM performance is verified using simulation results.
3. Perform circuit simulation.	3.1 <u>Circuit schematics</u> are developed using the appropriate simulation software. 3.2 Simulation parameters are configured according to circuit requirements. 3.3 <u>Circuit simulation</u> is executed to verify circuit operation. 3.4 Simulation waveforms and output results are analyzed. 3.5 Simulation results are verified against design specifications.
4. Operate measurement instruments safely.	4.1 <u>Measurement instruments</u> are selected according to testing requirements. 4.2 Instruments are connected and operated following

	safety procedures. 4.3 Voltage, current, and waveform measurements are performed correctly. 4.4 Measured results are recorded and compared with expected values. 4.5 Instruments are handled and stored according to workplace safety requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Power electronics principles and circuit fundamentals	1.1 Power Conversion Principles 1.2 Switching Principles 1.3 Energy Conversion 1.4 Circuit Analysis 1.5 Power Flow 1.6 Basic Electrical Laws
2. Power electronic components	2.1 Power Diodes 2.2 MOSFETs 2.3 IGBTs 2.4 Thyristors (SCRs) 2.5 Inductors 2.6 Capacitors
3. PWM principles and switching techniques	3.1 Pulse Width Modulation (PWM) 3.2 Hard Switching 3.3 Soft Switching 3.4 High-Frequency Switching 3.5 Unipolar Switching 3.6 Bipolar Switching
4. PWM control methods	4.1 Sinusoidal PWM (SPWM) 4.2 Space Vector PWM (SVPWM) 4.3 Hysteresis Current Control 4.4 Voltage Mode Control 4.5 Current Mode Control 4.6 Digital PWM Control
5. Circuit schematics	5.1 Circuit Diagrams 5.2 Schematic Capture 5.3 Electronic Components 5.4 Component Libraries 5.5 Netlists 5.6 Simulation Software
6. Circuit simulation	6.1 DC Analysis 6.2 AC Analysis 6.3 Transient Analysis 6.4 Steady-State Simulation

	6.5 Switching Simulation 6.6 Functional Verification
7. Measurement instruments	7.1 Digital Multimeter (DMM) 7.2 Oscilloscope 7.3 Clamp Meter 7.4 Function Generator 7.5 DC Power Supply 7.6 LCR Meter

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fundamentals of power electronics circuits and converter operation. 1.2 Power converter topologies, components, voltage, current, power, and efficiency concepts. 1.3 PWM principles, switching techniques, duty cycle, and control methods. 1.4 Power electronics circuit simulation methods and waveform analysis. 1.5 Selection and operation of electrical measurement instruments. 1.6 Safe measurement practices, data recording, and result verification.
2. Underpinning Skills	2.1 Understanding power electronics fundamentals and circuit principles 2.2 Selecting power electronic components according to circuit requirements 2.3 Explaining PWM control principles, waveforms, and switching techniques 2.4 Configuring and performing power electronics circuit simulation 2.5 Analyzing simulation waveforms and verifying circuit performance 2.6 Selecting and operating measurement instruments safely 2.7 Recording and evaluating electrical measurement results
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace

4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interfaced power stages with MCUs. 1.2 configured control signals. 1.3 implemented embedded control functions. 1.4 applied protection techniques.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY EMBEDDED CONTROL AND SENSOR INTEGRATION	Nominal Duration: 08 hrs.	Unit Code: SICIP-SCD-ASS-03-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Apply Embedded Control and Sensor Integration. It specifically includes the tasks of interfacing power stages with MCUs, configuring control signals, implementing embedded control functions and applying protection techniques.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interface power stages with MCUs.	1.1 <u>Power stages and microcontroller interfaces</u> are identified according to system requirements. 1.2 Input and output connections are configured between the MCU and power stage. 1.3 <u>Interface circuits</u> are verified for correct signal transmission. 1.4 Hardware connections are tested before system operation. 1.5 Interface performance is validated according to design requirements.
2. Configure control signals.	2.1 <u>Control signals</u> are configured according to system specifications. 2.2 PWM and digital control signals are generated for power stage operation. 2.3 <u>Signal timing and operating parameters</u> are verified before implementation. 2.4 Control signal outputs are tested using appropriate measurement instruments. 2.5 Signal performance is validated against design requirements.
3. Implement embedded control functions.	3.1 <u>Embedded control algorithms</u> are developed according to application requirements. 3.2 Sensor inputs are integrated into the control system. 3.3 Control functions are implemented using the selected microcontroller platform. 3.4 <u>Embedded software testing</u> is carried out to verify system operation. 3.5 Control performance is evaluated and optimized according to system requirements.
4. Apply protection techniques.	4.1 Protection requirements are identified according to system specifications.

	4.2 Protection methods are implemented.
	4.3 Protection circuits and control functions are tested under operating conditions.
	4.4 Fault conditions are detected and appropriate protective actions are verified.
	4.5 Protection performance is confirmed according to safety and design requirements.

Range of Variables

Variable	Range May include but not limited to:
1. Power stages and microcontroller interfaces	1.1 Half-Bridge Power Stage 1.2 Full-Bridge Power Stage 1.3 DC-DC Converter Stage 1.4 Microcontroller (MCU) 1.5 Digital Signal Controller (DSC) 1.6 Interface Circuits
2. Interface circuits	2.1 Level Shifters 2.2 Gate Driver Circuits 2.3 Signal Isolation Circuits 2.4 Optocouplers 2.5 Signal Conditioning Circuits 2.6 Interface Buffers
3. Control signals	3.1 PWM Signals 3.2 Digital Output Signals 3.3 Enable Signals 3.4 Reset Signals 3.5 Fault Signals 3.6 Feedback Signals
4. Signal timing and operating parameters	4.1 Duty Cycle 4.2 Switching Frequency 4.3 Dead Time 4.4 Pulse Width 4.5 Signal Delay 4.6 Timing Sequence
5. Embedded control algorithms	5.1 PID Control Algorithm 5.2 State Machine Algorithm 5.3 Motor Control Algorithm 5.4 MPPT Algorithm 5.5 Digital Filtering Algorithm 5.6 Protection and Fault Handling Algorithm
6. Embedded software testing	6.1 Functional Testing 6.2 Unit Testing 6.3 Integration Testing 6.4 Hardware-in-the-Loop (HIL) Testing

	6.5 Debugging Tools
	6.6 Simulation Results
7. Protection methods	7.1 Overcurrent Protection (OCP) 7.2 Overvoltage Protection (OVP) 7.3 Overtemperature Protection (OTP) 7.4 Short-Circuit Protection (SCP) 7.5 Undervoltage Lockout (UVLO) 7.6 Fault Detection Circuits

Curricular Content Guide

1. Underpinning Knowledge	1.1 Power stage and microcontroller interface concepts. 1.2 MCU input/output connections, signal transmission, and hardware integration. 1.3 PWM generation, digital control signals, and timing parameters. 1.4 Embedded control algorithms, sensor integration, and microcontroller-based control systems. 1.5 Protection techniques, fault detection, and safety control methods. 1.6 Testing, validation, and optimization of embedded power control systems.
2. Underpinning Skills	2.1 Interfacing power stages with microcontroller systems 2.2 Configuring MCU input and output connections for power control 2.3 Generating and controlling PWM and digital control signals 2.4 Validating signal timing and control parameters 2.5 Implementing embedded control algorithms and sensor integration 2.6 Testing and optimizing embedded control functions 2.7 Applying protection techniques for power electronic systems 2.8 Verifying fault detection and protective control performance
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector

	4.3 Stationery
	4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interfaced power stages with MCUs. 1.2 configured control signals. 1.3 implemented embedded control functions. 1.4 applied protection techniques.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM POWER SYSTEM DESIGN DOCUMENTATION	Nominal Duration: 08 hrs.	Unit Code: SICIP-SCD-ASS-04-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Prepare Power System Design Documentation. It specifically includes the tasks of preparing power electronic schematics, interpreting PCB layout requirements, preparing bills of materials (BOM) & preparing test documentation and reports.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare power electronic schematics.	1.1 <u>Circuit requirements</u> are interpreted according to the design specification. 1.2 Power electronic schematics are prepared using the appropriate EDA software. 1.3 Components and connections are verified for accuracy and completeness. 1.4 <u>Schematic documents</u> are reviewed against design requirements. 1.5 Final schematics are approved for PCB layout preparation.
2. Interpret PCB layout requirements.	2.1 <u>PCB layout requirements</u> are interpreted from design specifications. 2.2 Component placement, routing, and layer requirements are identified. 2.3 Power, ground, and thermal management considerations are reviewed. 2.4 PCB layout is checked for compliance with design rules. 2.5 Layout requirements are verified before PCB implementation.
3. Prepare bills of materials (BOM).	3.1 <u>Components</u> are identified according to the circuit design. 3.2 Component specifications and quantities are compiled into the BOM. 3.3 Manufacturer part numbers and technical details are verified. 3.4 BOM is reviewed for completeness and accuracy. 3.5 Final BOM is prepared according to organizational standards.
4. Prepare test documentation and reports.	4.1 <u>Test procedures</u> are prepared according to project requirements. 4.2 Test results and measurement data are recorded accurately.

	4.3 Test reports are prepared using the required format.
	4.4 Findings and observations are documented clearly.
	4.5 Final test documentation is reviewed and submitted according to organizational requirements.

Range of Variables

Variable	Range May include but not limited to:
1. Circuit requirements	1.1 Design Specifications 1.2 Functional Requirements 1.3 Electrical Requirements 1.4 Performance Requirements 1.5 Safety Requirements 1.6 Project Documentation
2. Schematic documents	2.1 Circuit Diagrams 2.2 Bill of Materials (BOM) 2.3 Netlist Report 2.4 Design Notes 2.5 Electrical Rule Check (ERC) Report 2.6 Design Review Checklist
3. PCB layout requirements	3.1 Design Specifications 3.2 PCB Design Rules 3.3 Board Dimensions 3.4 Layer Stack-Up 3.5 Manufacturing Requirements 3.6 Electrical Requirements
4. Components	4.1 Power Semiconductor Devices 4.2 Passive Components 4.3 Integrated Circuits (ICs) 4.4 Connectors 4.5 Protection Devices 4.6 Electromechanical Components
5. Test procedures	5.1 Test Plan 5.2 Functional Test 5.3 Performance Test 5.4 Electrical Test 5.5 Safety Test 5.6 Verification
6. Test reports	6.1 Functional Test Report 6.2 Performance Test Report 6.3 Measurement Report 6.4 Validation Report 6.5 Test Summary Report 6.6 Standard Report Format

Curricular Content Guide

1. Underpinning Knowledge	1.1 Power electronic circuit requirements and schematic development principles. 1.2 EDA tools for schematic preparation and design verification. 1.3 PCB layout fundamentals, component placement, routing, layers, and design rules. 1.4 Power PCB considerations, including grounding, thermal management, and safety requirements. 1.5 Bill of Materials (BOM) preparation, component specifications, and part identification. 1.6 Test procedure development, measurement data recording, and technical report preparation. 1.7 Design review, documentation standards, and project submission requirements.
2. Underpinning Skills	2.1 Interfacing power stages with microcontroller systems 2.2 Configuring MCU input and output connections for power control 2.3 Verifying hardware interface signals and system performance 2.4 Generating and controlling PWM and digital control signals 2.5 Implementing embedded control algorithms and sensor integration 2.6 Applying protection techniques for power electronic systems 2.7 Verifying fault detection and protective control performance
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared power electronic schematics. 1.2 interpreted PCB layout requirements. 1.3 prepared bills of materials (BOM). 1.4 prepared test documentation and reports.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation-specific Competencies

Unit of Competency: APPLY POWER SEMICONDUCTOR DEVICES AND SWITCHING TECHNIQUES	Nominal Duration: 32 hrs.	Unit Code: SICIP-SCD-ASS-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to Apply Power Semiconductor Devices and Switching Techniques. It specifically includes the tasks of identifying power semiconductor devices, analyzing switching characteristics and losses, selecting devices for safe efficient operation and simulating MOSFET and IGBT characteristics.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify power semiconductor devices.	1.1 <u>Power semiconductor devices</u> are identified according to application requirements. 1.2 Characteristics of diodes, MOSFETs, IGBTs, and thyristors are interpreted. 1.3 Device symbols, terminals, and operating modes are identified. 1.4 <u>Device specifications</u> are verified using manufacturer datasheets. 1.5 Appropriate devices are selected based on circuit requirements.
2. Analyze switching characteristics and losses.	2.1 <u>Switching characteristics</u> are analyzed according to device operating conditions. 2.2 Conduction and switching losses are calculated using device parameters. 2.3 <u>Switching waveforms</u> are interpreted to evaluate device performance. 2.4 Factors affecting switching efficiency are identified and analyzed. 2.5 Analysis results are verified against design requirements.
3. Select devices for safe and efficient operation.	3.1 <u>Device ratings</u> are evaluated according to circuit requirements. 3.2 Electrical, thermal, and switching characteristics are considered during device selection. 3.3 Safe operating limits are verified using manufacturer datasheets. 3.4 Selected devices are assessed for efficiency, reliability, and application suitability. 3.5 Device selection is confirmed according to design

	specifications.
4. Simulate MOSFET and IGBT characteristics.	4.1 MOSFET and IGBT simulation models are configured using the appropriate simulation software. 4.2 <u>Simulation parameters</u> are set according to circuit requirements. 4.3 Device simulations are performed to evaluate switching characteristics. 4.4 Simulation waveforms and performance results are analyzed. 4.5 Simulation results are verified against design specifications.

Range of Variables

Variable	Range May include but not limited to:
1. Power semiconductor devices	1.1 Power Diodes 1.2 Power MOSFETs 1.3 IGBTs 1.4 Thyristors (SCRs) 1.5 TRIACs 1.6 Fast Recovery Diodes
2. Device specifications	2.1 Manufacturer Datasheets 2.2 Maximum Ratings 2.3 Electrical Characteristics 2.4 Thermal Characteristics 2.5 Switching Parameters 2.6 Package Information
3. Switching characteristics	3.1 Turn-On Time 3.2 Turn-Off Time 3.3 Rise Time 3.4 Fall Time 3.5 Switching Frequency 3.6 Gate Drive Characteristics
4. Switching waveforms	4.1 Gate Voltage Waveform 4.2 Drain-Source Voltage Waveform 4.3 Collector-Emitter Voltage Waveform 4.4 Drain/Collector Current Waveform 4.5 Turn-On Waveform 4.6 Turn-Off Waveform
5. Device ratings	5.1 Voltage Rating 5.2 Current Rating 5.3 Power Rating 5.4 Frequency Rating 5.5 Surge Current Rating 5.6 Isolation Voltage Rating
6. Simulation parameters	6.1 Supply Voltage 6.2 Switching Frequency

	6.3 Gate Drive Voltage
	6.4 Load Conditions
	6.5 Temperature
	6.6 Time Step Settings

Curricular Content Guide

1. Underpinning knowledge	1.1 Types, symbols, terminals, and operating principles of power semiconductor devices. 1.2 Characteristics of diodes, MOSFETs, IGBTs, and thyristors. 1.3 Interpretation of manufacturer datasheets and device specifications. 1.4 Switching characteristics, conduction losses, and switching losses. 1.5 Factors affecting switching efficiency and device performance. 1.6 Electrical, thermal, and switching parameters for safe device selection. 1.7 MOSFET and IGBT simulation concepts, modeling, and waveform analysis. 1.8 Device evaluation based on efficiency, reliability, and application requirements.
2. Underpinning skills	2.1 Interpreting device symbols, terminals, and operating modes 2.2 Verifying device specifications using manufacturer datasheets 2.3 Analyzing switching characteristics and power losses 2.4 Evaluating switching efficiency and device performance 2.5 Assessing device ratings, reliability, and application suitability 2.6 Configuring and simulating MOSFET and IGBT characteristics 2.7 Analyzing and verifying simulation results against design specifications
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Tools & equipment 4.4 Job specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 identified power semiconductor devices. 1.2 analyzed switching characteristics and losses. 1.3 selected devices for safe and efficient operation. 1.4 simulated MOSFET and IGBT characteristics.
2. Methods of assessment	Methods of assessment should include: 2.1 Written Test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IMPLEMENT GATE DRIVE CIRCUITS AND THERMAL MANAGEMENT	Nominal Duration: 32 hrs.	Unit Code: SICIP-SCD-ASS-02-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to implement gate drive circuits and thermal management. It specifically includes the tasks of designing gate driver circuits, applying gate drive techniques, configuring protection and dead-time and selecting thermal management solutions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Design gate driver circuits.	1.1 <u>Gate driver requirements</u> are identified according to power device specifications. 1.2 Gate driver circuits are designed using appropriate design tools. 1.3 <u>Driver components</u> are selected according to voltage, current, and switching requirements. 1.4 Gate driver circuits are verified against design specifications. 1.5 Completed gate driver designs are prepared for implementation.
2. Apply gate drive techniques.	2.1 <u>Gate drive techniques</u> are applied according to power device requirements. 2.2 Gate resistance, bootstrap circuits, and switching parameters are configured. 2.3 <u>Gate drive signals</u> are verified to ensure correct device operation. 2.4 Switching performance is analyzed and optimized. 2.5 Gate drive implementation is validated according to design requirements.
3. Configure protection and dead-time.	3.1 Protection requirements are identified according to circuit specifications. 3.2 <u>Dead-time settings</u> are configured to prevent switching faults. 3.3 Overcurrent, overvoltage, and short-circuit protection functions are implemented. 3.4 Protection circuits are tested under operating conditions. 3.5 Protection and dead-time configurations are verified for safe operation.
4. Select thermal management solutions.	4.1 <u>Thermal requirements</u> are evaluated according to power dissipation levels. 4.2 Heat sinks, cooling methods, and thermal interface

	materials are selected. 4.3 <u>Thermal performance</u> is analyzed using design calculations or simulation results. 4.4 Thermal management solutions are verified to ensure safe operating temperatures. 4.5 Selected thermal management solutions are documented according to design requirements.
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Range of Variables

Variable	Range
	May include but not limited to:
1. Gate driver requirements	1.1 Power Device Specifications 1.2 Gate Drive Voltage 1.3 Gate Drive Current 1.4 Switching Frequency 1.5 Isolation Requirements 1.6 Design Specifications
2. Driver components	2.1 Gate Driver IC 2.2 Gate Resistor 2.3 Bootstrap Capacitor 2.4 Isolated Power Supply 2.5 Optocoupler/Digital Isolator 2.6 Protection Components
3. Gate drive techniques	3.1 High-Side Gate Drive 3.2 Low-Side Gate Drive 3.3 Isolated Gate Drive 3.4 Bootstrap Gate Drive 3.5 Push-Pull Gate Drive
4. Gate drive signals	4.1 PWM Signal 4.2 Gate Pulse 4.3 High-Side Drive Signal 4.4 Low-Side Drive Signal 4.5 Gate Voltage Waveform 4.6 Switching Timing
5. Dead-time settings	5.1 Dead-Time Controller 5.2 Dead-Time Interval 5.3 PWM Controller 5.4 Gate Driver Configuration 5.5 Switching Frequency 5.6 Complementary Switching Signals
6. Thermal requirements	6.1 Power Dissipation 6.2 Junction Temperature 6.3 Ambient Temperature 6.4 Thermal Resistance 6.5 Operating Temperature Range 6.6 Device Power Rating

7. Thermal performance	7.1	Thermal Calculations
	7.2	Thermal Simulation
	7.3	Temperature Distribution
	7.4	Junction Temperature Analysis
	7.5	Heat Flow Analysis
	7.6	Thermal Resistance Network

Curricular Content Guide

1. Underpinning knowledge	1.1 Gate driver circuit principles and requirements for power semiconductor devices. 1.2 Gate drive techniques, gate resistance, bootstrap circuits, and switching control methods. 1.3 Switching performance analysis and optimization techniques. 1.4 Protection methods, including overcurrent, overvoltage, short-circuit protection, and dead-time control. 1.5 Thermal management concepts, heat dissipation, heat sinks, and cooling techniques. 1.6 Thermal calculations, simulation methods, and safe operating temperature requirements. 1.7 Gate driver and protection circuit testing and validation procedures.
2. Underpinning skills	2.1 Designing gate driver circuits for power semiconductor devices 2.2 Applying gate drive techniques for efficient device operation 2.3 Configuring gate resistance, bootstrap circuits, and switching parameters 2.4 Analyzing and optimizing gate drive performance 2.5 Configuring protection circuits and dead-time control 2.6 Implementing overcurrent, overvoltage, and short-circuit protection 2.7 Analyzing and verifying thermal performance for safe operation
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual)

	4.2 Specifications and work instructions
	4.3 Learning materials
	4.4 Tools and equipment

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 designed gate driver circuits. 1.2 applied gate drive techniques. 1.3 configured protection and dead-time. 1.4 selected thermal management solutions.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DESIGN POWER CONVERSION SYSTEMS	Nominal Duration: 40 hrs.	Unit Code: SICIP-SCD-ASS-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to Design Power Conversion Systems. It specifically includes the tasks of developing DC-DC converter topologies, explaining inverter and UPS systems, selecting magnetic and filter components and verifying converter performance.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Develop DC-DC converter topologies.	1.1 <u>DC-DC converter requirements</u> are interpreted according to system specifications. 1.2 <u>Suitable converter topologies</u> are selected based on application requirements. 1.3 Converter circuits are designed using appropriate design and simulation tools. 1.4 <u>Design parameters</u> are verified through simulation and analysis. 1.5 Converter design is validated against project requirements.
2. Explain inverter and UPS systems.	2.1 <u>Inverter and UPS operating principles</u> are interpreted according to system requirements. 2.2 System components and operating modes are identified. 2.3 <u>Power conversion processes</u> are explained using system block diagrams. 2.4 Inverter and UPS applications are related to industrial requirements. 2.5 System operation is verified using simulation or design documentation.
3. Select magnetic and filter components.	3.1 <u>Magnetic components</u> are selected according to converter design requirements. 3.2 Inductors, transformers, capacitors, and filters are evaluated for circuit performance. 3.3 <u>Component ratings</u> are verified using design calculations and manufacturer datasheets. 3.4 Selected components are assessed for efficiency, thermal performance, and reliability. 3.5 Component selection is confirmed according to design specifications.
4. Verify converter performance.	4.1 Converter performance is evaluated using simulation and test results.

	4.2 Output voltage, current, efficiency, and ripple characteristics are analyzed. 4.3 <u>Performance issues</u> are identified and corrective actions are applied. 4.4 Converter operation is verified against design specifications. 4.5 Final performance results are documented according to project requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. DC-DC converter requirements	1.1 Input Voltage 1.2 Output Voltage 1.3 Output Current 1.4 Power Rating 1.5 Efficiency Requirements 1.6 Load Requirements
2. Suitable converter topologies	2.1 Buck Converter 2.2 Boost Converter 2.3 Buck-Boost Converter 2.4 Flyback Converter 2.5 Forward Converter 2.6 Full-Bridge Converter
3. Design parameters	3.1 Voltage Regulation 3.2 Current Ripple 3.3 Switching Frequency 3.4 Conversion Efficiency 3.5 Output Voltage Ripple 3.6 Simulation Results
4. Inverter and UPS operating principles	4.1 DC-AC Power Conversion 4.2 Pulse Width Modulation (PWM) 4.3 Standby UPS 4.4 Line-Interactive UPS 4.5 Online (Double Conversion) UPS 4.6 Power Backup Operation
5. Power conversion processes	5.1 AC to DC Conversion 5.2 DC to AC Conversion 5.3 Battery Charging 5.4 Battery Discharging 5.5 Power Flow Block Diagram 5.6 Control Circuit Operation

6. Magnetic components	6.1 Inductors 6.2 Transformers 6.3 Ferrite Cores 6.4 Common-Mode Chokes 6.5 Differential-Mode Chokes 6.6 Output Filters
7. Component ratings	7.1 Inductance 7.2 Capacitance 7.3 Current Rating 7.4 Voltage Rating 7.5 Power Rating 7.6 Manufacturer Datasheets
8. Performance issues	8.1 Voltage Regulation Errors 8.2 Excessive Ripple 8.3 Low Efficiency 8.4 Thermal Issues

Curricular Content Guide

1. Underpinning knowledge	1.1 DC-DC converter topologies, operating principles, and design requirements. 1.2 Converter circuit design, simulation, and performance analysis methods. 1.3 Inverter and UPS systems, components, operating modes, and applications. 1.4 Magnetic components, including inductors, transformers, capacitors, and filters. 1.5 Component selection based on electrical ratings, efficiency, thermal performance, and reliability. 1.6 Converter performance parameters, including voltage, current, efficiency, and ripple analysis. 1.7 Testing, validation, and documentation of power converter performance.
2. Underpinning skills	2.1 Developing DC-DC converter topologies 2.2 Selecting converter topologies based on application requirements 2.3 Designing and simulating DC-DC converter circuits 2.4 Selecting magnetic components and filter circuits 2.5 Evaluating component ratings, efficiency, and reliability 2.6 Analyzing converter performance using simulation and test results 2.7 Verifying converter performance against design specifications
3. Underpinning attitudes	3.1 Commitment to occupational health and safety

	3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	The Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 developed DC-DC converter topologies. 1.2 explained inverter and UPS systems. 1.3 selected magnetic and filter components. 1.4 verified converter performance.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INTEGRATE CONTROL, MOTOR DRIVE AND RENEWABLE ENERGY SYSTEMS	Nominal Duration: 36 hrs.	Unit Code: SICIP-SCD-ASS-04-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to integrate control, motor drive and renewable energy systems. It specifically includes the tasks of interfacing power stages with MCUs, implementing motor control functions, integrating BMS, solar inverter and MPPT systems and validating industrial control applications.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interface power stages with MCUs.	1.1 <u>Power stages and microcontroller interfaces</u> are identified according to system requirements. 1.2 Hardware connections and control signals are configured between the MCU and power stage. 1.3 Interface circuits are tested to verify proper communication and operation. 1.4 <u>Signal timing and control functions</u> are validated before system integration. 1.5 Interface performance is verified according to design specifications.
2. Implement motor control functions.	2.1 <u>Motor control requirements</u> are interpreted according to application specifications. 2.2 Motor control algorithms and PWM signals are implemented using the selected controller. 2.3 <u>Motor drive parameters</u> are configured and tested. 2.4 Motor operation is monitored and optimized for stable performance. 2.5 Motor control functions are verified against design requirements.
3. Integrate BMS, solar inverter and MPPT systems.	3.1 BMS, solar inverter, and MPPT system requirements are identified according to application needs. 3.2 <u>System components</u> are connected and configured following design specifications. 3.3 Communication and control functions are verified between integrated subsystems. 3.4 <u>Integrated system performance</u> is evaluated under operating conditions. 3.5 System integration is validated according to project requirements.
4. Validate industrial control applications.	4.1 <u>Industrial control systems</u> are tested according to application requirements. 4.2 System performance and operating parameters are

	measured and analyzed. 4.3 Functional and safety requirements are verified during system operation. 4.4 Identified issues are corrected and system performance is revalidated. 4.5 Validation results are documented according to organizational standards.
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Range of Variables

Variable	Range May include but not limited to:
1. Power stages and microcontroller interfaces	1.1 Half-Bridge Power Stage 1.2 Full-Bridge Power Stage 1.3 DC-DC Converter Power Stage 1.4 Microcontroller (MCU) 1.5 Digital Signal Controller (DSC) 1.6 Interface Circuits
2. Signal timing and control functions	2.1 PWM Timing 2.2 Dead-Time Control 2.3 Switching Sequence 2.4 Interrupt Handling 2.5 ADC Sampling Timing 2.6 Synchronization Signals
3. Motor control requirements	3.1 Application Specifications 3.2 Speed Control Requirements 3.3 Torque Control Requirements 3.4 Direction Control Requirements 3.5 Load Requirements 3.6 Motor Performance Specifications
4. Motor drive parameters	4.1 PWM Frequency 4.2 Duty Cycle 4.3 Motor Speed 4.4 Motor Current 4.5 Voltage Reference 4.6 Acceleration and Deceleration Settings
5. System components	5.1 Battery Pack 5.2 PV Modules 5.3 Inverter Module 5.4 MPPT Controller 5.5 Charge Controller 5.6 Power Conversion Stage
6. Integrated system performance	6.1 Power Conversion Efficiency 6.2 MPPT Tracking Performance

	6.3 Battery Charging Performance 6.4 System Stability 6.5 Load Performance 6.6 Operating Temperature
7. Industrial control systems	7.1 Motor Drive Systems 7.2 Variable Frequency Drives (VFDs) 7.3 Programmable Logic Controllers (PLCs) 7.4 Industrial Power Converters 7.5 Renewable Energy Systems 7.6 Automation Control Systems

Curricular Content Guide

1. Underpinning knowledge	1.1 Power stage and microcontroller interfacing concepts. 1.2 MCU control signals, communication, timing, and hardware integration methods. 1.3 Motor control principles, PWM techniques, and control algorithms. 1.4 Battery Management System (BMS), solar inverter, and MPPT system fundamentals. 1.5 Industrial control system integration and performance evaluation. 1.6 System testing, safety validation, troubleshooting, and technical documentation.
2. Underpinning skills	2.1 Interfacing power stages with microcontroller systems 2.2 Configuring hardware connections and control signals 2.3 Testing and validating MCU interface performance 2.4 Implementing motor control functions and PWM algorithms 2.5 Configuring and optimizing motor drive performance 2.6 Verifying communication and control between integrated subsystems 2.7 Validating industrial control system performance and safety 2.8 Analyzing and documenting system validation results
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interfaced power stages with MCUs. 1.2 implemented motor control functions. 1.3 integrated BMS, solar inverter and MPPT systems. 1.4 validated industrial control applications.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM CIRCUIT DESIGN, PCB DEVELOPMENT AND TROUBLESHOOTING	Nominal Duration: 32 hrs.	Unit Code: SICIP-SCD-ASS-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to perform circuit, PCB development and Troubleshooting. It specifically includes the tasks of preparing power electronics schematics, developing PCB layouts, assembling and prototype circuits, diagnosing circuit faults and mitigating EMI/EMC issues.		

Elements and Performance Criteria

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare power electronic schematics.	1.1 <u>Circuit requirements</u> are interpreted according to the design specification. 1.2 Power electronic schematics are prepared using the appropriate EDA software. 1.3 Components and circuit connections are verified for accuracy and completeness. 1.4 Schematic design is reviewed against project requirements. 1.5 Final schematics are approved for PCB layout development.
2. Develop PCB layouts.	2.1 <u>PCB layout requirements</u> are interpreted according to the schematic design. 2.2 Component placement and PCB routing are completed following design rules. 2.3 Power, ground, thermal, and EMI considerations are applied during PCB layout. 2.4 PCB layout is verified using design rule checks (DRC). 2.5 Completed PCB layouts are prepared for fabrication or prototyping.
3. Assemble and prototype circuits.	3.1 Components are selected and assembled according to the PCB design. 3.2 <u>Prototype circuits</u> are constructed following assembly procedures. 3.3 Electrical connections are inspected before power-up. 3.4 Prototype circuits are tested to verify functional operation. 3.5 Assembly results are documented according to project requirements.
4. Diagnose circuit faults.	4.1 Circuit faults are identified using appropriate test equipment. 4.2 Measurement results and circuit behavior are analyzed

	to determine fault causes. 4.3 Faults are corrected according to troubleshooting procedures. 4.4 <u>Repaired circuits</u> are retested to verify proper operation. 4.5 Troubleshooting results are documented according to organizational requirements.
5. Mitigate EMI/EMC issues.	5.1 <u>EMI/EMC requirements</u> are identified according to applicable standards. 5.2 Sources of electromagnetic interference are analyzed during circuit operation. 5.3 PCB layout, grounding, shielding, and filtering techniques are applied to reduce EMI/EMC issues. 5.4 Circuit performance is verified after implementing EMI/EMC improvements. 5.5 EMI/EMC mitigation results are documented according to project requirements.

Range of Variables

Variable	Range May include but not limited to:
1. Circuit requirements	1.1 Design Specifications 1.2 Functional Requirements 1.3 Electrical Requirements 1.4 Safety Requirements 1.5 Performance Requirements 1.6 Project Documentation
2. PCB layout requirements	2.1 PCB Design Rules 2.2 Layer Stack-up 2.3 Board Dimensions 2.4 Component Footprints 2.5 Manufacturing Constraints 2.6 Layout Specifications
3. Prototype circuits	3.1 PCB Assembly 3.2 Breadboard Prototype 3.3 Soldered Prototype 3.4 Wiring 3.5 Mechanical Assembly 3.6 Assembly Procedures
4. Repaired circuits	4.1 Functional Testing 4.2 Power-Up Testing 4.3 Signal Measurement 4.4 Load Testing 4.5 Performance Testing

	4.6 Test Equipment
5. EMI/EMC requirements	5.1 EMC Standards 5.2 EMI Standards 5.3 Regulatory Requirements 5.4 Compliance Specifications 5.5 Design Guidelines 5.6 Project Requirements

Curricular Content Guide

1. Underpinning knowledge	1.1 Power electronic schematic design principles and EDA tools. 1.2 PCB layout fundamentals, design rules, component placement, and routing techniques. 1.3 Power PCB considerations, including grounding, thermal management, and EMI control. 1.4 PCB assembly, prototyping procedures, and circuit testing methods. 1.5 Circuit fault diagnosis, troubleshooting techniques, and measurement tools. 1.6 EMI/EMC principles, interference sources, shielding, filtering, and mitigation techniques. 1.7 PCB verification, prototype validation, and technical documentation practices.
2. Underpinning skills	2.1 Preparing power electronic schematics using EDA tools 2.2 Developing PCB layouts following design rules 2.3 Applying power, grounding, thermal, and EMI considerations 2.4 Verifying PCB layouts using design rule checks (DRC) 2.5 Assembling and prototyping power electronic circuits 2.6 Diagnosing and troubleshooting circuit faults 2.7 Mitigating EMI/EMC issues using appropriate design techniques 2.8 Verifying and documenting PCB and circuit performance
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual)

	4.2 Learning materials
	4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 prepared power electronic schematics. 1.2 developed PCB layouts. 1.3 assembled and prototype circuits. 1.4 diagnosed circuit faults. 1.5 mitigated EMI/EMC issues.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DEVELOP SEMICONDUCTOR SYSTEMS INTEGRATION PROJECT	Nominal Duration: 28 hrs.	Unit Code: SICIP-SCD-ASS-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to develop semiconductor systems integration project. It specifically includes the tasks of explaining semiconductor packaging and reliability, planning the system integration project, developing and optimizing the prototype, testing system performance and preparing technical documentation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain semiconductor packaging and reliability.	1.1 <u>Semiconductor packaging types and functions</u> are identified according to application requirements. 1.2 Packaging technologies and reliability factors are interpreted. 1.3 Thermal, mechanical, and environmental considerations are identified for semiconductor devices. 1.4 <u>Packaging and reliability requirements</u> are related to system performance. 1.5 Packaging and reliability information is verified using technical documentation.
2. Plan the system integration project.	2.1 Project requirements and system specifications are reviewed before implementation. 2.2 System architecture, components, and integration activities are planned. 2.3 Project resources, milestones, and implementation schedule are prepared. 2.4 <u>Required hardware, software, and test equipment</u> are identified and organized. 2.5 Project plan is reviewed and approved before development.
3. Develop and optimize the prototype.	3.1 <u>System prototype</u> is developed according to the approved design. 3.2 Hardware and software components are integrated and configured. 3.3 Prototype performance is evaluated under operating conditions. 3.4 <u>Design modifications</u> are implemented to improve system performance and reliability. 3.5 Optimized prototype is verified against project requirements.
4. Test system	4.1 System functionality is tested according to the test plan.

performance.	4.2 Electrical, thermal, and operational performance are measured and analyzed. 4.3 <u>System faults and performance issues</u> are identified and corrected. 4.4 Test results are verified against design specifications. 4.5 Final system performance is validated before project completion.
5. Prepare technical documentation.	5.1 Project reports, design files, test results, and supporting documents are compiled. 5.2 System design, implementation, and testing procedures are documented clearly. 5.3 Technical drawings, schematics, and bills of materials are reviewed for accuracy. 5.4 <u>Project documentation</u> is organized according to organizational standards. 5.5 Final technical documentation is submitted according to project requirements.

Range of Variables

Variable	Range May include but not limited to:
1. Semiconductor packaging types and functions	1.1 Dual In-line Package (DIP) 1.2 Small Outline Integrated Circuit (SOIC) 1.3 Quad Flat Package (QFP) 1.4 Ball Grid Array (BGA) 1.5 Quad Flat No-Lead (QFN) 1.6 Chip Scale Package (CSP)
2. Packaging and reliability requirements	2.1 Thermal Performance 2.2 Mechanical Reliability 2.3 Electrical Performance 2.4 Signal Integrity 2.5 Product Lifetime 2.6 Application Requirements
3. Required hardware, software, and test equipment	3.1 Development Boards 3.2 Electronic Components 3.3 EDA Software 3.4 Programming Tools 3.5 Test Instruments 3.6 Laboratory Equipment
4. System prototype	4.1 Hardware Prototype 4.2 Software Prototype 4.3 Embedded System Prototype 4.4 Integrated System Prototype 4.5 Prototype Design

	4.6 Development Platform
5. Design modifications	5.1 Hardware Optimization 5.2 Software Optimization 5.3 Circuit Improvements 5.4 Performance Tuning 5.5 Reliability Enhancements 5.6 Design Corrections
6. System faults and performance issues	6.1 Hardware Faults 6.2 Software/Firmware Errors 6.3 Communication Failures 6.4 Thermal Issues 6.5 Performance Degradation 6.6 Corrective Actions
7. Project documentation	7.1 Final Project Report 7.2 Technical Manual 7.3 User Manual 7.4 Test and Validation Report 7.5 Submission Package 7.6 Approval Records

Curricular Content Guide

1. Underpinning knowledge	1.1 Semiconductor packaging types, technologies, and reliability factors. 1.2 Thermal, mechanical, and environmental considerations in semiconductor systems. 1.3 System integration project planning, architecture, resources, and scheduling. 1.4 Hardware and software integration, prototype development, and optimization techniques. 1.5 System testing methods, performance measurement, and fault analysis. 1.6 Electrical, thermal, and operational validation of integrated systems. 1.7 Technical documentation, design files, test reports, schematics, and BOM preparation.
2. Underpinning skills	2.1 Explaining semiconductor packaging and reliability concepts 2.2 Planning semiconductor system integration projects 2.3 Preparing system architecture, resources, and implementation schedules 2.4 Developing and optimizing system prototypes 2.5 Testing and validating system performance 2.6 Analyzing and resolving system faults and performance

	issues 2.7 Preparing and managing technical documentation and project deliverables
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 explained semiconductor packaging and reliability. 1.2 planned the system integration project. 1.3 developed and optimized the prototype. 1.4 tested system performance. 1.5 prepared technical documentation.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Applied Semiconductor Systems: Power Devices and Industrial Integration
Number of Trainees:	25

Course-wise Training Space:

- Computer Lab : 400 sft
- Workshop : 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Computer/Laptops	01
2.	Printer	01
3.	Multimedia projector	01
4.	Power Electronics and Semiconductor Device Simulation Software (SPICE, LTspice, PLECS, PSIM, MATLAB/Simulink or Equivalent)	25
5.	Semiconductor Device Characterization Software	25
6.	Linux/Windows Operating System	25
7.	Central File/Storage Server	01
8.	Gigabit Network Switch (48-Port)	01
9.	Enterprise Router	01
10.	Digital Storage Oscilloscope (100–200 MHz)	10
11.	Programmable DC Power Supply	10
12.	Function/Arbitrary Waveform Generator	10
13.	Digital Multimeter	25
14.	LCR Meter	05
15.	Power Analyzer	02
16.	Semiconductor Curve Tracer / Device Analyzer	01
17.	Thermal Imaging Camera	01
18.	Electronic Load	05
19.	Soldering & Rework Station	10
20.	PCB Prototyping Tools and Accessories	05
21.	Power Semiconductor Training Kits (MOSFET, IGBT, SiC, GaN Evaluation Boards)	10

22.	Gate Driver Development Kits	10
23.	DC-DC Converter and Inverter Training Kits	10
24.	Motor Drive and Industrial Control Training Kits	05
25.	Heat Sink and Thermal Management Demonstration Kits	10

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Applied Semiconductor Systems: Power Devices and Industrial Integration, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.